



Association of Zoos & Aquariums

2020 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2020 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **16 March 2020**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 16 March 2020, **regardless of cause**, will NOT be eligible for consideration. NO DEADLINE EXCEPTIONS WILL BE MADE.*

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2020 and funds will be available 1 October 2020. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The CGF application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: CGFapp@aza.org. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. *Only* completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

CGF Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled “**Application**” and attached to your CGF Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.

Note that “spell checker” and “track changes” functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their Project Abstract and Project Proposal in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.*

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list. Include a project abstract no longer than 1,500 characters, including spaces.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
 - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
 - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your CGF Application email:

- ☒ **Curricula Vitae.** A curriculum vitae (CV) is required for **each** PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed a **maximum** length of two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, “**CV Investigator’s Last Name.**” *CVs that exceed the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **EACH** CI institution *and* collaborating institution must be attached to the CGF Application email as separate documents titled “**SIS**

Institution's Name. The *Statement of Institutional Support* form can be found with the rest of the CGF Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions, or AZA Animal Programs may submit such items as appendices.

Note that letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form. In addition, letters of support/endorsement **from AZA Animal (Taxon Advisory Groups, Species Survival Plans®, etc.) and SAFE Programs** are not accepted although coordination with AZA Animal and SAFE Programs is encouraged and should be described in Question 1. To assure Animal and SAFE Program review of all relevant CGF submissions, AZA Program Leaders will serve as expert reviewers in the CGF application review process and may express their support for a proposal at that time.

- ☒ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your CGF Application email. Each permit attached to the CGF Application email must be titled **"Permit #."** *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2020 to ensure award eligibility.*
- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for CGF funding and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the CGF Application email. This attachment must be submitted as a Microsoft Word document titled **"Resubmission Form"** and attached to your CGF Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the CGF Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your CGF Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Project Proposal should be included in a separate document titled **"Citations"** and attached to the CGF Application email.
- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos or publications. Appendices must be attached to the CGF Application email titled **"Appendix #."**

Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA's website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2020 CGF Application and related forms via a **single email, with the proposal title as the subject line**, to CGFapp@aza.org no later than 5:00 p.m. EDT on 16 March 2020. Hard copies of any application materials or applications received after this deadline will not be accepted *regardless of cause*.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2020 CGF Application email **must** include the following attachments

- CGF Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution; and
- A Proposal Resubmission form, if applicable.
- The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, s/he should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the CGF Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 7 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

2020 AZA CONSERVATION GRANTS FUND
Cover Sheet

Project Number: [REDACTED]
For AZA use only

Title of Project: Intersecting human land-use, livelihoods, and conservation of the endangered Zanzibar red colobus

Amount Requested: \$30,878 (round to the nearest whole dollar)

Project Start Date: January 2021

End Date: December 2021

(CGF funding is for one year only, beginning no earlier than the award date)

Principal Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. Principal Investigator (PI) Name and Title: [REDACTED]

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (required; available at: <http://www.aza.org/cgf-information-for-applicants/>) ☒

Individual AZA Member Number: [REDACTED] (required)

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: [REDACTED]

Phone: [REDACTED]

E-mail: [REDACTED]

2. Co-Investigator (CI) Name and Title:

CV Attached? ☐ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at: <http://www.aza.org/cgf-information-for-applicants/>) ☐ (required)

Individual AZA Member Number: [REDACTED] (required)

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☐ (required)

Institutional Address: [REDACTED]

Phone: [REDACTED]

E-mail: [REDACTED]

3. Co-Investigator (CI) Name and Title:

CV Attached? ☐ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at: <http://www.aza.org/cgf-information-for-applicants/>) ☐ (required)

Individual AZA Member Number: [REDACTED] (required)

Institution/Organization: [REDACTED]

Statement of Institutional Support Form Attached? ☐ (required)

Institutional Address: [REDACTED]

Phone: [REDACTED]

E-mail: [REDACTED]

Note: If additional Co-Investigators need to be included, add their information in a separate document titled “**Additional Co-Investigators**” and include it as an attachment to your CGF Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the PI’s and CI’s institutions listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the CGF Application email.

Collaborating Institution Names:

University Statement of Institutional Support Form Attached? ☒ (required)

Statement of Institutional Support Form Attached? ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your CGF Application email.

Project Categories (choose two that best describe the project):

☐ Animal/Population Management (*Ex situ*)

☒ Research

☐ Conservation Education

☐ Animal Health

☒ Field Conservation (*In situ*)

☐ Animal Welfare

Project Abstract (limit 1,500 characters with spaces):

Red colobus monkeys are among the most threatened groups of primates in the world, with more than half of the taxa classified as either Endangered or Critically Endangered on the IUCN Red List of Threatened Species. One of these monkeys is the endangered Zanzibar red colobus (ZRC), an endemic species found in the Zanzibar Archipelago, part of the United Republic of Tanzania. ZRC are facing an increasingly uncertain future due to the combination of habitat loss, habitat fragmentation, hunting, and the environmental impacts of climate change. This project will address critical needs for ZRC conservation by producing ZRC genetic data and human behavioral data, which will then be combined to directly inform conservation management recommendations. By utilizing a landscape genetics approach, our data will allow us to determine how ZRC migration across the landscape is affected by different anthropogenic landscape features. The human data will illuminate determinants of human land-use and flexibility of livelihoods in ZRC areas, particularly within the framework of climate change, both as it is being experienced in the present and how it may alter livelihoods and land-use patterns in the future. These two results will then be combined to create a spatially explicit recommendation for land-use planning that maintains ZRC genetic variation and adaptive potential while also strengthening resilience in human livelihoods to climate change.

2020 AZA CONSERVATION GRANTS FUND

Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance for AZA animal/conservation/education programs or institutions, and/or how it reflects the priorities of the AZA Board (e.g., sustainability of managed populations, SAFE: Savings Animals From Extinction) or of related Animal or SAFE species programs. **LIMIT 5,000 CHARACTERS with spaces**

The Zanzibar red colobus monkey (ZRC) is an endangered species endemic to the Zanzibar Archipelago (United Republic of Tanzania), which is part of the 'Coastal Forests of Eastern Africa' biodiversity hotspot [1]. The ZRC is native to Unguja Island (1,670 km²) and has a current population size of 5,862 individuals, which are distributed across roughly 21 sites in a fragmented manner [2]. ZRC are facing an increasingly uncertain future due to a combination of habitat loss, habitat fragmentation, hunting, and climate change [3]. A long history of economic reliance on agriculture, a recent sharp reliance on tourism, and steep human population growth has led to intense resource use and land conversion, with rates of deforestation on Zanzibar reaching ~10 km² per annum [4, 5]. This has left the ZRC to navigate a highly heterogeneous landscape with a suboptimal agricultural matrix between its preferred forest habitats. Approximately 31% of the total population exists outside of protected areas and some of these sites are very isolated. We currently lack knowledge of how landscape discontinuities impact ZRC dispersal between forest patches and thus gene flow, which is crucial for the maintenance of genetic diversity in populations. Genetic diversity can act as a buffer against stochastic processes, and populations with low genetic diversity have been shown to be more at risk to extinction via disease outbreak and environmental change [6, 7, 8]. Understanding the current dispersal patterns of ZRC is thus crucial to maintaining adaptive potential and building effective population management.

Informed population management of the ZRC also requires incorporation of climate change. Several studies have established that climate change has already caused quantifiable shifts in sea levels, patterns of precipitation, and soil quality on Unguja [9, 10, 11]. The ZRC will not only be directly impacted by these environmental and phenological shifts as a forest dependent species with a folivorous diet [12], but they will also face indirect pressure due to the effects of climate change on human livelihoods in ZRC areas. This is especially problematic as poverty is a persistent issue in Zanzibar, with 30% of people living on less than \$1.90 USD a day. Economic instability creates a strong reliance on the land and natural resources, while environmental changes create uncertainty in these practices. These issues likely feedback on one another, resulting in highly vulnerable human livelihoods and an increased level of dependence on the land. ZRC share habitats and resources with humans across their range, meaning their long-term viability as a population is also reliant on humans having stable livelihoods. This makes understanding the determinants of human livelihoods and how resilient they are to climate change an important element of creating effective conservation strategies for ZRC.

The overarching goal of this research is to support long-term survival of the ZRC population, while also building resilience to climate change into human livelihoods. The specific questions addressed will be: (1) how do specific types of land-use impact gene flow in ZRC? (2) what are the determinants of different types of human land-use near ZRC habitats and how might these change in the future? and (3) how flexible are people's livelihoods in ZRC habitat areas and can changes be made to enhance ZRC migration? This project will combine spatially explicit ZRC biological data and human behavioral data to create management recommendations that prioritize the most viable ZRC subpopulations for conservation while simultaneously suggesting land-use schemes which encourage ZRC migration and resilience in human livelihoods. Such land-use planning is a critical need for the future survival of the ZRC and the well-being of people on Zanzibar.

Unguja is extremely vulnerable to climate change as a small low lying island state, tying this research directly to AZA's climate change position statement. AZA also prioritizes research related to species and habitat conservation. The ZRC is an endangered species endemic to a biodiversity hotspot, as well as the flagship species for Zanzibar and central to ecotourism critical to sustainable development plans aimed at poverty relief. Lastly, the AZA CGF aims to empower zoo professionals to be leaders in their field. The PI has nearly a decade of experience working in AZA accredited facilities, including five-years as a primate zookeeper at the [REDACTED] Zoo. This provided the PI with experience directly applicable to the proposed project, but more importantly inspired her to pursue her doctorate and actively contribute to primate conservation. The PI strongly believes that ex-situ and in-situ partnerships strengthen conservation initiatives, and would

like to one day take her personal knowledge and experiences back to the AZA community through employment after completion of her doctorate.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

The long-term goal of this research is to ensure the survival and viability of the ZRC. This directly relates to the goal of the proposed project, which is the creation of a spatially explicit land-use plan that simultaneously ensures the long-term viability of the ZRC and supports human livelihoods by building resilience to climate change. This will be achieved by understanding how land-use influences human livelihoods and alters patterns of ZRC movement, which will then allow for landscape planning which takes both into account. At the end of the research, management recommendations will be distributed to the appropriate governmental and public stakeholders in order to support local decision making. To accomplish the project goal, the following objectives will be completed:

Objective 1: Determine how land-use on Unguja impacts movement and migration of ZRC through a landscape genetics approach. This will allow for the identification of landscape features that influence genetic connectivity, including which specific land-use types and crops are more permeable to ZRC dispersal. In order to complete this objective, 1) non-invasive (i.e. fecal) georeferenced samples will be collected across forests containing ZRC groups (~21 sites, ~300 samples), 2) samples will be genotyped, 3) GIS data will be collected for the island, 4) population genetic analyses will be conducted to identify how ZRC genetic variation is structured across the island, 5) landscape genetics analyses will be conducted to identify which landscape features impede ZRC movement and which ones allow for ZRC gene flow.

Objective 2: Identify the determinants of human land use in ZRC areas, including how flexible people are in their agricultural practices.

Semi-structured interviews will be conducted to generate information concerning agricultural practices and land tenure, human-wildlife interactions, socioeconomic factors, and climate change. In order to complete this objective, (1) potential participants will be identified from red colobus habitat adjacent communities through stratified selection using land-use types and distance to natural forest patches as proxies for potential human-colobus interaction, (2) key informant interviews and quantitative household surveys will be carried out using Participant Rural Appraisal tools, (3) transcripts from interviews will be analyzed by identifying common themes and collating related statements, (4) household surveys and appropriate data from interviews will be coded as binary, ordinal, and continuous data, and (5) spatially explicit social data models will be created. Knowing how and why people are using the land will enable us to recommend land-use schemes that encourage ZRC migration, are politically and socially tenable, and build resilience in the human community to climate change.

Objective 3: Creation of spatially explicit management recommendations for ZRC.

The results from Objectives 1 and 2 will be combined to create conservation recommendations which maintain adaptive potential and genetic diversity for the ZRC and are also socially and politically tenable. In order to meet this objective we will, (1) employ geospatial software to overlay the landscape genetics and social data, (2) utilize geospatial analysis techniques, such as network analysis, to assist in prioritizing regions for conservation action, (3) once regions are selected, we will revisit the original data in order to determine what type of land-use scheme (i.e., shifts in agricultural practices or crop types) best suit those specific regions, and (4) once recommendations have been finalized, they will be compiled into a document for dissemination.

This project has an estimated total timeline of 15 months. The portion of the project a CGF grant would fund is the first 12 months of this period. This includes a field season followed by lab work, analyses, and creation of conservation recommendations. Dissemination of results will occur during the three month period after CGF support ends. The outputs from the 12 months of work funded by a CGF grant would include all of the results from the analyzed ZRC genetic and human social data as well as the conservation recommendations. In the long-term, the conservation management recommendations created by this project will strengthen viability of the ZRC population across its range.

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and proposed actions are appropriate for addressing the conservation issue addressed. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

In order to address the question (1) how do specific types of land-use impact gene flow in ZRC, a landscape genetics approach will be taken. This will utilize standard methodology for sample collection, microsatellite genotyping, and analysis [13]. Non-invasive fecal samples will be collected in the field and stored in tubes with 6 mL of RNAlater in order to stabilize DNA. Accompanying GPS tags will be marked for each sample. The PI will accomplish this by following ZRC groups at a distance; there will be no direct contact with any animals during sampling and personal protective equipment will be worn to protect both the PI and the animals. The sampling scheme is based on a population census completed in 2015, which estimates there are 342 social groups at roughly 21 sites across the island [2]. 300 samples will be collected across the island, roughly one sample per social group, in order to create a continuous sampling distribution [14, 15]. The PI will attempt to sample as many different social groups as possible, but in the event not all social groups can be located, locales with a higher density of social groups will be sampled more intensely. The PI has experience carrying out fieldwork of this exact setup with wild chacma baboons (*Papio ursinus*) in South Africa [16], and collaborators [REDACTED] and Zoe Melvin have been collecting ZRC fecal samples for hormonal analysis since 2018, showing the feasibility of this approach. A permit is required from the Revolutionary Government of Zanzibar's Department of Forestry and Non-Renewable Natural Resources to collect samples from the colobus. Collaborator [REDACTED] will aid in obtaining permits and has successfully secured permits for non-invasive colobus sampling in the past.

Samples will be exported to the United States for DNA extraction and analysis. Sample preparation and DNA extraction will be conducted in the [REDACTED] Lab at the University [REDACTED]. This project will use well-tested and vetted methodology to genotype microsatellites from fecal-extracted DNA. We will genotype each sample at 15 microsatellite loci, as microsatellites are highly variable and it has been shown that as few as 8-12 loci can be sufficient to distinguish related individuals with statistical confidence in a landscape genetics study [17]. Previously developed microsatellites are available for colobus monkeys, which enables us to carry out this research without needing to develop our own markers [18]. In order to bring the costs of the total lab analysis down, we will use multiplex PCR to reduce the overall number of reactions that need to be carried out. The samples will then be genotyped using an ABI 3730 capillary DNA sequencer at [REDACTED] University's Center [REDACTED]. Multi-locus genotypes will be combined with remotely sensed geospatial data for Unguja. Geospatial analysis software, such as ArcMap, will be utilized for these analyses. For this project, isolation-by-distance (IBD) is the null hypothesis. IBD assumes that genetic difference is best explained by geographic distance. Our operating hypothesis is that genetic difference will be better explained by anthropogenically created landscape features than it can be by distance alone. In order to test this hypothesis we will use the isolation-by-resistance (IBR) model and circuit theory [19]. Isolation-by-resistance predicts a positive relationship between genetic and resistance distances, a theoretic distance metric based on circuit theory. Circuit theory processes landscapes as if they were conductive surfaces, with electrons flowing across a circuit to model genes moving across the landscape [20]. Circuit theory allows one to consider all possible paths between two samples when assessing landscape influence on gene flow. Circuit-based connectivity will be calculated using the software Circuitscape 4.0. In order to do this, we will first develop resistance surfaces for the different landscape features we want to test as barriers to ZRC movement across the landscape. Resistance surfaces will be created utilizing remotely sensed data and will be parameterized using the R package ResistanceGA [21]. The output from this process will be a model of ZRC gene flow across the island, which features hinder ZRC gene flow, and which anthropogenic land-uses or crop types are most permeable or impermeable to ZRC dispersal.

In order to address the questions (2) what are the determinants of different types of human land-use near ZRC habitats and how might these change in the future, and (3) how flexible are people's livelihoods in ZRC habitat areas and can changes be made to enhance ZRC migration, we will employ social science methodologies. Many lines of questioning will be nested within these broader questions in order to understand agricultural practices, land tenure, human-wildlife interactions, socioeconomic factors, and climate change. A mixed-methods approach will be carried out, combining informant interviews, which utilize Participatory Rural Appraisal tools [22], and quantitative household surveys. Contacts will be made and the PI will set up interviews in one-on-one and qualitative focus group settings. We are most interested in eliciting information from farmers, but we may decide there are other key informants, such as individuals working in non-governmental conservation organizations, important to interview. Participants will be identified from red colobus habitat adjacent communities through stratified selection based on land-use types and distance to natural forest patches as proxies for potential human-colobus interaction [23]. All information collected will be done so in a way that preserves the participants' anonymity. The PI has intermediate proficiency in Swahili and is continuing to develop her language skills through coursework at the University [REDACTED] so that she can perform the interviews herself.

Transcripts from interviews will be analyzed by identifying common themes and collating related statements [24]. Household surveys, which comprise our socioeconomic data, will be coded as binary, ordinal, and continuous data as appropriate. We will also quantitatively code any appropriate data from our interviews, such as categorical coding of crop types or binary coding of subsistence versus commercial agriculture [25]. This will allow us to create spatially explicit social data models. We will take this information and utilize geospatial processing software, such as ArcMap, to create raster and vector models which display continuously and discretely distributed data across the landscape. Once we have these data layers, we will be able to overlay the social data layers onto the landscape genetics analysis, producing a

biologically and socially informed spatially explicit model. This will allow us to create spatially explicit recommendations for land-use planning that incorporates the needs of both the ZRC and human livelihoods.

4. If previous CGF funding was obtained for this or a related project, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

N/A

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

This entirety of this project will take roughly 15 months to complete, beginning with fieldwork and ending with the dissemination of findings and recommendations to stakeholders in Zanzibar. The one year portion supported by a CGF grant would include field work, lab work, and data analysis and is detailed below.

- January 2021 – July 2021 (6 months): The field season will be focused on non-invasive genetic sampling across Unguja with the help of field assistants, as well as conducting semi-structure interviews with farmers living in colobus habitat areas.

- July 2021 - December 2021 (6 months): Lab work to genotype samples from the 2nd field season. Landscape genetics analyses will then be completed utilizing the genetic data and geospatial data. Data from the semi-structured interviews will be analyzed utilizing mixed-methods. Conservation recommendations will be created by combining the ZRC genetic and human social data.

Additionally, we are very aware of the uncertainty that the current COVID-19 outbreak presents. The timeline of this project has already been shifted due to current restrictions in travel, if however there are still restrictions on travel come winter 2021, we imagine our entire timeline could be shifted by 3-6 months, with the proposed work pushed back to a summer 2021 start date.

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

██████████ is a PhD student at ██████████, and the proposed project represents her doctoral research. ██████████ has her MSc in Conservation Biology from the ██████████, where her research focused on using immunogenetics to inform conservation management of chacma baboons. ██████████ has a wide variety of field and lab experience, as well as over a decade of experience working at AZA accredited institutions, giving her a broad perspective of in ex-situ and in-situ primate conservation. ██████████ will be the researcher collecting all of the data in the field, will do all of the lab work and data analysis, and will create the subsequent conservation recommendations and disseminate them.

██████████ (Collaborator) is an Associate Professor at ██████████ and specializes in the use of molecular genetics to aid in the conservation of endangered species. His research has focused on colobus monkeys, and he is a lead editor of the ██████████. ██████████ aided in the development of the proposed research and will oversee all aspects of the project including data collection in the field, analyzing and interpreting results, and the creation and dissemination of conservation recommendations.

██████████ (Collaborator) is a Lecturer at ██████████ and Director of the ██████████, which studies how the endangered ZRC is coping with human-induced changes to their habitat. ██████████ has provided feedback on project design and will assist with on-the-ground research logistics in Zanzibar, as well as the creation of conservation recommendations based on these data.

██████████ (Collaborator) is the ██████████ Director for ██████████ in Africa and the former ██████████ Director. He has worked across Africa and is involved in research, protected area design and management, climate change adaptation, and community conservation initiatives. ██████████ has provided essential feedback on project design, will be a source for on-the-ground logistics in Zanzibar, and will also assist with the creation of conservation recommendations and communications with the Zanzibar government.

██████████ (Collaborator) is a PhD student at ██████████. She has studied ZRC groups in multiple locations across Unguja since 2018, validating non-invasive collection of fecal samples. ██████████ has provided feedback on project design and will assist with sample collection and logistics in the field.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed). **LIMIT 3,000 CHARACTERS with spaces**

We will monitor the progress of the proposed research over the course of the 12 month funding period. We will use a process indicator system, making sure actions are complete in a timely manner (see proposed timeline). The actions we will monitor for progress and completion in each objective are:

Objective 1: Determine how land-use on Unguja impacts movement and migration of ZRC through a landscape genetics approach. 1) non-invasive (i.e. fecal) georeferenced sample collection across forests containing ZRC groups (~21 sites, ~300 samples), 2) sample genotyping 3) GIS data collection, 4) population genetic analyses 5) landscape genetics analyses

Objective 2: Identify the determinants of human land use in ZRC areas, including how flexible people are in their agricultural practices. (1) identification of potential participants from red colobus habitat adjacent communities, (2) completion of key informant interviews and quantitative household surveys using Participant Rural Appraisal, (3) analysis of transcripts from interviews (4) coding of household surveys and appropriate data from interviews as binary, ordinal, and continuous data, and (5) creation of spatially explicit social data rasters.

Objective 3: Creation of spatially explicit management recommendations for ZRC. (1) overlay of landscape genetics output with social data models in geospatial analysis software, (2) determination of which regions should be prioritized for conservation action in order to support dispersal of red colobus, (3) determination of which areas human livelihoods are most vulnerable, (4) creation of recommendations that combine this information, and (5) delivery of recommendations to stakeholders.

We will evaluate the impact of the research following project completion for at least 2 years. Specifically, we will identify which of the recommendations we make are converted to policy, how that policy is implemented, whether it encourages ZRC movement across the landscape, and how it is received by people living adjacent to ZRC habitat.

8. Explain and identify the means, and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

The Red Colobus Conservation Action Plan under the IUCN/SSC Primate Specialist Group calls for the integration of local communities into red colobus conservation through bottom up approaches, and for the identification and prioritization of ZRC conservation units in order to strengthen conservation in and around protected areas. Completion of the proposed research would assist in simultaneously addressing these objectives by identifying requirements necessary to ensure a sustainable future for humans and the ZRC. The results from the proposed work will be disseminated directly by the PI and collaborators to relevant stakeholders (e.g., Department of Forestry and Non-Renewable Natural resources, communities in ZRC habitat areas) in the 3 month period after CGF support ends, as well as via publication in peer-reviewed journals, presentations at conferences, and as a doctoral dissertation. Written reports will be given to the IUCN/SSC Primate Specialist Group, as the IUCN 'research needed' section calls for data on ZRC threats, area-based management plans, population trends, and habitat trends. The PI will also engage in science communication throughout the project, utilizing social media platforms to share information and research experiences with the general public. The PI believes making these results available and approachable for both local and international stakeholders is key to engendering support not just for ZRC conservation, but for wildlife conservation in general.

9. Have all the necessary permits been obtained for the project?

a) ☐ Yes ☒ No ☐ Not Applicable

If yes, list the permits here and attach one copy of each to your CGF Application email. Attachments should be titled "**Permit #.**"

N/A

- b) If the necessary permits have not yet been obtained, please explain why. Pending permits must be obtained and submitted to AZA by 15 June 2020.

We have not yet acquired permits as fieldwork is not starting until January 2021. Our research group, the [REDACTED] [REDACTED] has successfully secured permits for non-invasive Zanzibar red colobus sampling in the past.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

N/A

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

N/A

12. **Past CGF Support.** If any of the proposal's Principal or Co-Investigators listed on the Cover Sheet have received previous CGF support, complete the required information below for each.

- | | | | |
|----|---|------------------------------|-----------------------------|
| a) | Name of Investigator: N/A | Year of Award: N/A | Amount of Award: N/A |
| | Project Title: N/A | | |
| | Were all required reports (annual/final) submitted? | ☐ Yes | ☐ No |
| b) | Name of Investigator: N/A | Year of Award: N/A | Amount of Award: N/A |
| | Project Title: N/A | | |
| | Were all required reports (annual/final) submitted? | ☐ Yes | ☐ No |

Note: If additional awards need to be included, add their information to a separate document titled "Past CGF Projects" and include it as an attachment with your CGF Application email.

2020 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF does not fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation/management/education goals.

CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's CGF Scientific Advisory Committee may recommend that the CGF award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your CGF Application email.

Budget Items Form:

Budget Item	Method of Calculation (e.g., 2 persons x 10 days x \$25/day)	CGF- Requested Amount	Non-CGF Requested Amount (if applicable)	Date Expected from Non-CGF Source (if applicable and not yet received)	Name of Non-CGF Source (e.g., Grant Fund, Matching)	Total Expense (CGF+Non-CGF)
Air Fare	1 person x 1 roundtrip ticket from [REDACTED] to Zanzibar	\$2,100	\$2,100	July 2020	[REDACTED]	\$2,100
Visa	1 person x 1 visitor visa for United Republic of Tanzania	\$100	\$100	July 2020	[REDACTED]	\$100
Field Assistants	3 field assistants x 100 days x \$15/day	\$0	\$4,500	May 2020	[REDACTED] of [REDACTED]	\$4,500
Jozani-Chwaka Bay National Park fee	1 person x 1 park entry fee for six months	\$200	\$200	July 2020	[REDACTED]	\$200
Field Lodgings	1 person x 186 days x \$12/day	\$1,860	\$372	May 2020	[REDACTED] of [REDACTED]	\$2,232
Car Rental	1 car x 30 days x \$30/day	\$900	\$900	July 2020	[REDACTED]	\$900
RNA later for sample collection	2 500ml bottles x \$454/bottle	\$908	\$908	July 2020	[REDACTED]	\$908
Driver's Permit	1 person x 1 temporary driver's permit for Zanzibar	\$10	\$10	July 2020	[REDACTED]	\$10
DNA Extraction	7 QIAGEN fecal DNA extraction kits x \$328/kit	\$2,300	\$2,300	July 2020	[REDACTED] und, Margot Marsh	\$2,300

[illegible]

BUDGET JUSTIFICATION

State why each item listed in the CGF Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

\$5k of the proposed lab and field work is being funded by [REDACTED], with additional funds being sought from the [REDACTED] Fund (submitted) [REDACTED] Fund (submitted), [REDACTED] c. The full project budget is laid out in this proposal, as we have not yet received notification about funding from any other agencies. Because of this, there is overlap in the requests we have made from other agencies and the AZA CGF. The intent is to combine any financial support received from the AZA CGF with funds from other grants to meet the full budget for the project. As soon as notification of funding from other agencies is received, the PI will send an updated budget to the AZA CGF.

Funding from this grant will cover costs associated with a mix of field work, lab work, and analyses. Six months of field work will be conducted in Zanzibar in order to collect non-invasive genetic samples from Zanzibar red colobus as well as to conduct interviews with people and collect social science data. Air fare and a visitor's visa are required for the PI to get to and enter Zanzibar, and field lodgings (6 months) will be necessary once in country. Non-invasive genetic sampling requires RNAlater (2 bottles at \$454 bottle; roughly 6ml used per sample) to stabilize DNA until the samples can be exported back to the US for DNA extraction. In addition, field assistants will be hired to help locate colobus and collect samples (funded by [REDACTED]). National Park fees, car rental (4 weeks at \$30/day), and a temporary driver's permit are all budget items necessary to complete both the genetics and social science components of the proposed work. The remaining budget items are all associated with lab work, which will take place at the [REDACTED]. Extraction kits are required to extract DNA from the fecal samples. 300 samples will be processed in total, which requires 7 QIAGEN fecal DNA extraction kits (7 kits at \$328/kit). Genotyping at 15 loci per sample, with 6 multiplexed reactions and 3 repeats per sample, will require 6 QIAGEN Multiplex PCR kits (\$2,500/kit). The samples will then be genotyped using an AB 3730 capillary DNA sequencer at [REDACTED] Center for [REDACTED]. 57 plates of genotyping (96 wells each) are needed for all of our samples (\$132/plate). The combined costs associated with the full project total roughly \$30,878.

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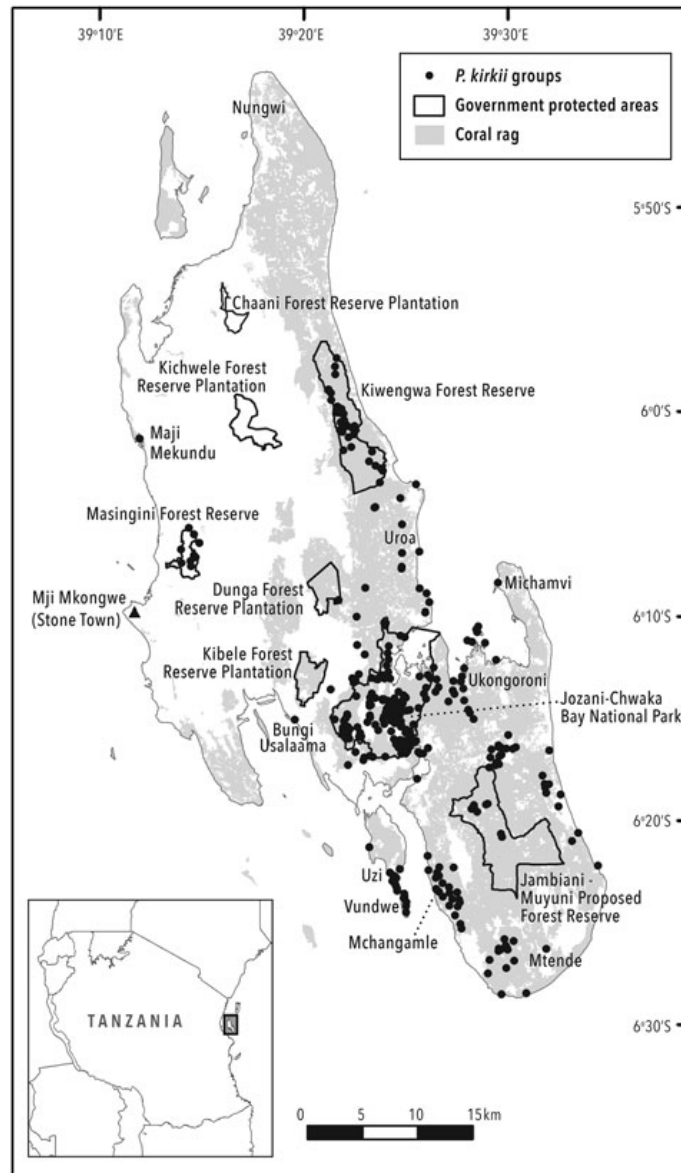
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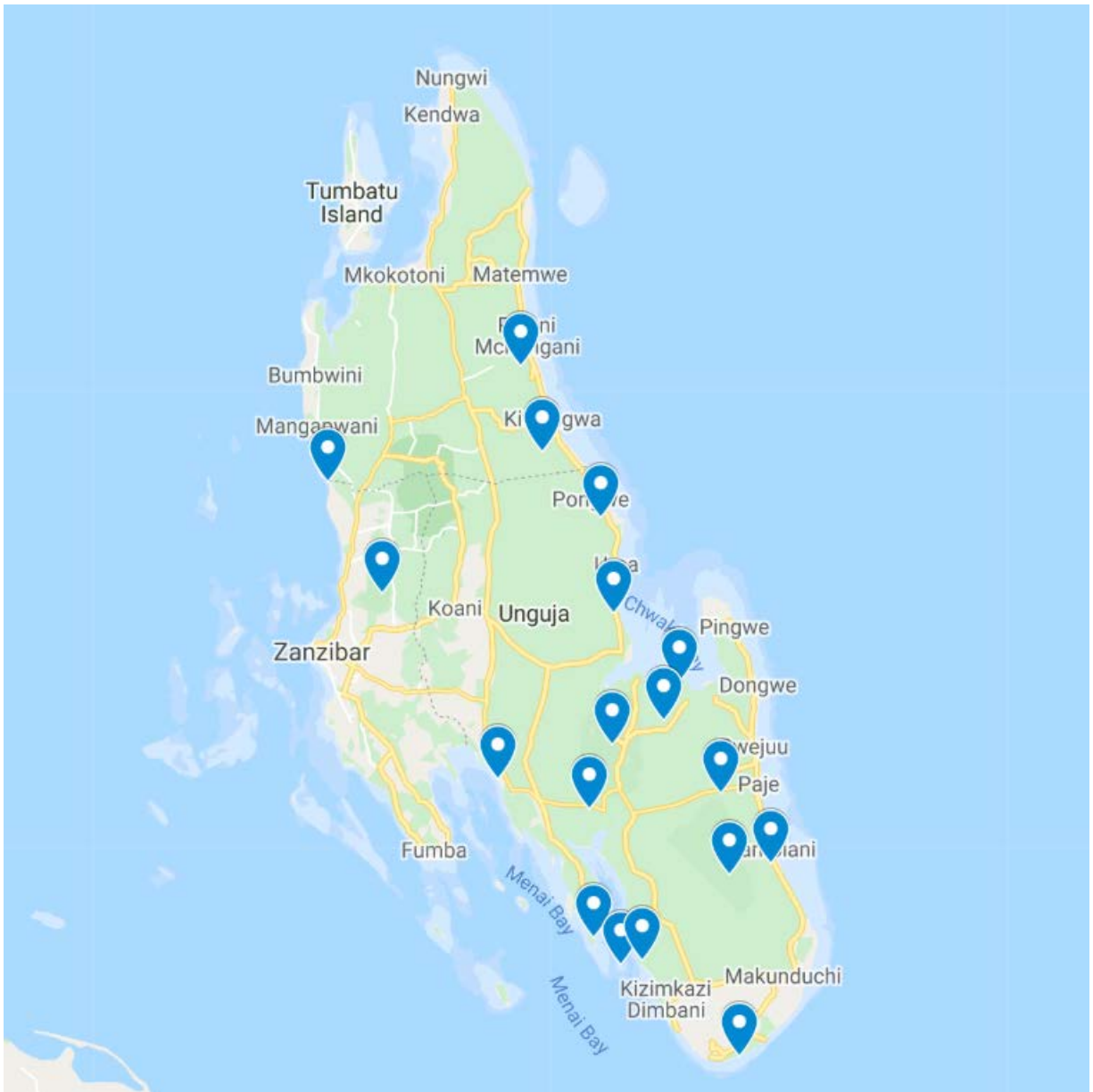
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Map 1: This project will take place on Unguja, the largest island of the Zanzibar Archipelago, United Republic of Tanzania, Africa.



Map 2: This map shows the locations of all Zanzibar red colobus (*Piliocolobus kirkii*) groups located and counted during a census carried out in 2015 on the island of Unguja, Zanzibar (Davenport et al. 2019).



Map 3: Non-invasive fecal sampling of Zanzibar red colobus will be carried out across Unguja. These sampling locales have been chosen based on Davenport et al.'s 2015 island-wide census of the Zanzibar red colobus population.

